

Barnabás Gellér

Software Engineer |
Industrial & Research Software |
Architecture & Integration

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CORE SKILLS

.NET | C# | JavaScript | TypeScript |
React | Blazor | WPF | REST APIs |
gRPC | SQL | PostgreSQL |
MongoDB | SQLite | CI/CD | Docker |
Azure DevOps | SonarQube | Git |
LLM-assisted development

DOMAINS

Industrial Automation
SCADA / HMI
ERP
Finance Software
Software Testing
Biomedical Research Software

LANGUAGES

Hungarian: Native or bilingual
English: Full professional
German: Limited working
Swedish: Elementary
Spanish: Elementary

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals
Automation Engineer Level 2
Automation Specialist Level 2

SUMMARY

With 8 years in technology, spanning software engineering and computer science education, I work where system design and implementation meet; turning unclear requirements, legacy constraints, and cross-system integrations into maintainable software that teams can understand, deliver, and support with confidence.

My technical foundation is .NET, C#, and application architecture, with practical work across JavaScript, React, Blazor, WPF, databases, APIs, CI/CD, production support, and industrial interfaces.

Across industrial automation, ERP, finance, software testing, and biomedical research software, I combine systems thinking with pragmatic creativity. I clarify actors, constraints, lifecycles, data flows, failure modes, and operational realities before choosing abstractions, and favor explicit boundaries, configurable behavior, and reversible change.

I use LLM tooling for implementation, code analysis, refactoring, documentation, planning, and faster feedback loops. I am open-minded about AI-assisted and agentic development where it improves speed and quality, while remaining deliberate about architecture, review, maintainability, and engineering ownership.

EXPERIENCE

Advanced Engineering Industrie Automation GmbH

August 2023 - Present
Hallein, Salzburg, Austria

SCADA Software Developer (Tech Lead)

Line-control and HMI software for PCB and semiconductor production, where my role combines architecture, implementation, integration work, delivery support, and technical direction.

- Technical owner for Comandra, a configurable SCADA/HMI and line-control platform built to support production-line behavior, customer-specific workflows, and maintainable runtime deployments.
- Designed integration structures between HMI screens, backend services, PLCs, device interfaces, databases, scripting, and host systems.
- Built and modernized production HMI applications across React, Blazor, and WPF frontend technologies.
- Created workflow, driver, localization, and scripting concepts that kept customer-specific behavior configurable instead of hardcoded.
- Improved engineering workflows around builds, deployments, documentation, and development feedback loops using Docker, self-hosted agents, proof builds, and LLM-assisted repository work.
- Delivered and supported five production-line projects, including modernizations, maintained systems, and a full rewrite of a multi-machine sorting-line application.
- Guided contributors in a small software team while working closely with PLC developers, customers, and engineering stakeholders.

Core technologies: .NET | C# | JavaScript | TypeScript | React | Blazor | WPF | gRPC | REST APIs | SQLite | MongoDB | TwinCAT | OPC UA | SECS/GEM | Docker | CI/CD | Git | GitHub Copilot | LLM-assisted development

Kyndryl

August 2022 - July 2023

Software Developer Team Lead

Hungary

Enterprise software modernization for an internal banking platform, where my role focused on architecture cleanup, frontend/backend separation, delivery quality, and roadmap alignment.

- Worked on the Salkku application transformation, gradually separating a legacy Razor Pages application into a React frontend and .NET API-based backend.
- Reworked dependency management by replacing legacy UnityContainer usage with proper dependency injection patterns.
- Introduced feature toggles to support step-by-step migration between embedded legacy pages and newly developed React pages.
- Improved engineering quality through Azure DevOps pipelines, SonarQube analysis, proof builds, code coverage, unit testing, integration testing, and frontend snapshot testing.
- Worked with Cucumber-based BDD practices, xUnit testing, existing DDD concepts, and messaging components involving NServiceBus and RabbitMQ.
- Built and communicated a two-year technical roadmap, planned quarterly milestones, and helped align technical priorities with management expectations.
- Supported coordination across the legacy and modernization teams through documentation, knowledge-sharing sessions, workshops, and responsibility clarification.

Core technologies: .NET | C# | Razor Pages | React | TypeScript | Azure DevOps | SonarQube | xUnit | Cucumber | BDD | RabbitMQ | NServiceBus | CI/CD | Feature Toggles | Dependency Injection

Tricentis

February 2022 - July 2022

Software Engineer

Hungary

Low-code software testing platform work, where my role focused on modernization, authentication integration, test coverage, and safe refactoring of enterprise product components.

- Migrated .NET Framework projects and package dependencies toward .NET Core.
- Integrated OAuth 2.0 and Azure AD for external API connections.
- Increased confidence in change through unit testing, regression testing, and code coverage improvements.
- Maintained user-data consistency during refactoring and modernization work.
- Worked in an environment where product stability, testability, and controlled change were central engineering concerns.

Core technologies: .NET Framework | .NET Core | C# | OAuth 2.0 | Azure AD | Azure DevOps | Unit Testing | Regression Testing | Code Coverage | Jira | Confluence

Danubisoft Kft.

December 2019 - February 2022

ERP Software Developer

Pécs, Hungary

ERP and warehouse-management software for automotive and industrial customers, where my role combined module development, data modeling, integrations, customer support, and deployment ownership.

- Designed and developed cost forecasting, data collection, and quality-control modules in a long-running ERP platform.
- Designed data models and API integrations for forecasting and external invoicing workflows.
- Built a full synchronization flow between the ERP platform and the Vectory invoicing system, covering product data, prices, orders, and invoices.
- Worked with PostgreSQL schema design, query optimization, production maintenance, customer support, bug fixing, and deployments.
- Developed customer-specific customization mechanisms using Lua-based script injection.
- Worked with AMQP-based communication and template-driven document generation.

Core technologies: C++ | PostgreSQL | AMQP | Lua | ERP Systems | Warehouse Management | System Integration | Query Optimization | Customer Support | Deployment

PTE Deák Pécs

August 2019 - June 2020

Teacher of Mathematics and Informatics

Pécs, Hungary

Mathematics and computer science teaching role focused on exam preparation, practical IT activities, and helping students build structured problem-solving habits.

- Prepared students for computer science exams, supported mathematics education, and developed extracurricular IT activities.

Core technologies: Mathematics | Computer Science Education | Exam Preparation | Problem Solving | Teaching

Arduino

October 2018 - May 2019

Educational Content Specialist

Malmö, Sweden

Technical education content for C++, electronics, embedded software, and beginner-friendly IT learning, where my role focused on clarity, curriculum fit, and technical accuracy.

- Revised and developed learning materials for C++, IT, electronics, and embedded software topics.
- Aligned educational content with different curricula and learner levels.
- Reviewed certification questions and learning materials for clarity, correctness, and practical relevance.

Core technologies: C++ | Technical Education | Embedded Software Basics | Electronics | Curriculum Alignment | Certification Review | Learning Material Development

Mathematics and informatics teaching role focused on e-learning material development, exam preparation, and beginner-friendly programming education.

- Developed e-learning materials and specialized mathematics curriculum.
- Prepared students for mathematics exams and supported structured problem-solving skills.
- Introduced programming concepts through Scratch, CodeCombat, and custom learning tools.

Core technologies: Mathematics | Informatics Education | E-Learning | Curriculum Development | Programming Basics | Problem Solving

PROJECTS

Comandra: Unified HMI and Line Controller Platform

April 2024 - Present

Associated with Advanced Engineering Industrie Automation GmbH

Configurable SCADA/HMI and line-control platform for PCB and semiconductor production, focused on unifying runtime behavior, operator interfaces, integrations, and deployment workflows.

- Re-architected separated HMI and line-controller concepts into a cohesive platform with a unified runtime and deployment model.
- Designed configurable workflow and scripting concepts to support line-specific behavior without hardcoding each customer setup.
- Built integration structures across HMI screens, backend services, PLC communication, device interfaces, databases, and host systems.
- Delivered the platform from architecture and implementation to rollout, deployment support, and operational hardening.

Core technologies: .NET | C# | Blazor | JavaScript | SQLite | TwinCAT | SECS/GEM | Docker

Kamra: Household Product Management System

October 2022 - Present

Open-source household stock management project for tracking pantry items, expiry dates, desired stock levels, and shopping needs, built as a practical AI-assisted development experiment.

- Designed the system around product tracking, stock levels, expiry awareness, and automatic shopping-list generation.
- Built product-data collection and shop-crawling concepts to support future enrichment, comparison, and automation features.
- Used the project as a playground for AI-assisted implementation, repository-driven documentation, and GitHub workflow automation.
- Deployed the application through a modern TypeScript-based stack with automated development and delivery workflows.

Core technologies: Angular | TypeScript | Node.js | GitHub Actions | Vercel | Web Crawling | AI-assisted development

Salkku: Banking Application Transformation

September 2022 - July 2023

Associated with Kyndryl

Modernization project for Salkku, a business-critical banking application at Samlink, focused on gradually separating legacy UI and backend responsibilities while preserving existing business logic and operational stability.

- Built and communicated a technical roadmap with quarterly milestones, priorities, and modernization steps.
- Helped split a legacy Razor Pages / MVC-style application into a React frontend and .NET API backend.
- Supported business logic preservation, authentication continuity, and controlled migration through feature toggles.
- Improved development quality through documentation, code and environment change tracking, testing practices, and pipeline-based quality checks.
- Coordinated a developer team of four through responsibility clarification, knowledge-sharing sessions, weekly discussions, and technical workshops.

Core technologies: .NET | C# | Razor Pages | React | TypeScript | Azure DevOps | SonarQube | xUnit | Cucumber | RabbitMQ | NServiceBus | Feature Toggles

EDUCATION

Eötvös Loránd University

2015 - 2018

Master's Degree, Teacher of Mathematics and Informatics

Focused on mathematics and informatics education, with practical work around ICT-supported learning, e-learning design, digital competencies, and gamification.

Core areas: Mathematics Education | Informatics Education | ICT Tools | E-Learning | Digital Competencies | Gamification

Eötvös Loránd University

2011 - 2015

Bachelor's Degree, Mathematician

Focused on mathematics, with advanced studies in combinatorics and geometry, and a 50-credit supplementary field in informatics.

Core areas: Mathematics | Combinatorics | Geometry | Informatics | Analytical Problem Solving

PUBLICATIONS

- International Informatic Challenge in Hungary
- Informatikai kompetenciák fejlesztése modern eszközökkel
- A háromszögek középpontok baricentrikus koordinátái